Architecture 1: reserve-pool put spread, fixed duration buckets, full backing.

Version 0.1 draft Chain Stellar / Soroban Settlement asset USDC

Audience engineering, risk, integration partners

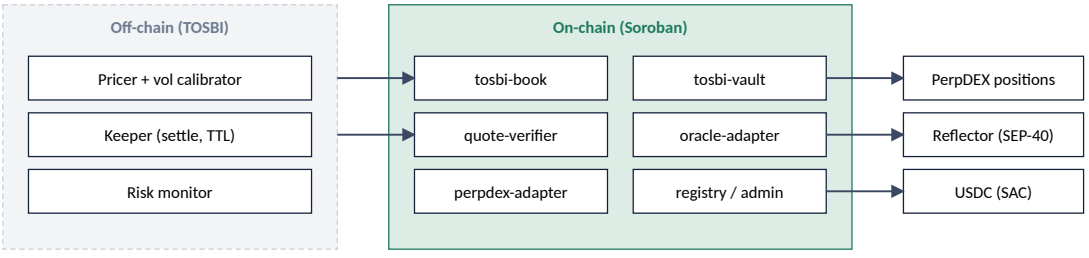
1. Scope
2. Notation
3. Product terms
4. Pricing
5. Capital rules
6. Settlement
7. Anti-abuse rules
8. Soroban contract architecture
9. Off-chain services
10. Initial risk parameters
11. Worked example
12. Monitoring
13. Test and validation plan
14. Roadmap after v0.1
15. References

§0 Scope

This document specifies the MVP protection product: a loss-linked, capped protection contract attached to a PerpDEX position, priced off-chain from a volatility table, reserved and settled on-chain, and fully backed by PerpDEX-supplied Protection Capital.

It covers product terms, pricing formulas, capital rules, settlement rules, anti-abuse rules, Soroban contract architecture, off-chain services, initial risk parameters, a worked example, monitoring, a test plan, and references.

Out of scope for v0.1: streaming/perpetual protection (Architecture 3), tranched vaults (Architecture 4), delta-hedging (Architecture 5).



Pricing happens off-chain and is signed; money, reservations, caps and settlement are enforced on-chain. Externals are the PerpDEX's position contract, the oracle, and the USDC token.

§1 Notation

Symbol	Meaning	Unit
`A`	Protected Amount, chosen by trader	USDC
`P_start`	Protection Start price (position mark at quote time)	USDC per unit
`Q = A / P_start`	Protected quantity	asset units
`R`	Coverage Range, chosen by trader	fraction (e.g. 0.25)
`P_end`	Coverage End price	USDC per unit
`D`	Duration	seconds
`τ = D / 31,557,600`	Duration in years (365.25-day year)	years
`σ_dir(asset, τ)`	Effective volatility from the vol table, per direction: `σ_down` prices LONG protection, `σ_up` prices SHORT protection	annualized fraction
`MaxPayout`	Maximum payout, equals reserved capital	USDC
`Π_fair`, `Π`	Fair (unloaded) premium; charged premium	USDC
`F`	Protection Fee (TOSBI revenue)	USDC
`K`	Protection Capital in vault	USDC
`U`	Utilization = Σ active MaxPayout / K	fraction
`C_{a,d}`	Concentration share of (asset a, direction d) = Σ MaxPayout(a,d) / K	fraction
`P_settle`	Settlement price	USDC per unit
`W(D)`	Settlement TWAP window	seconds
`N(·)`	Standard normal CDF	

Direction `d ∈ {LONG, SHORT}` and asset `a` are read from the PerpDEX Position Contract, never entered by the trader.

§2 Product terms

2.1 Trader chooses

- `A` Protected Amount, `A_min ≤ A ≤ min(A_max, position notional)`
- `D` Duration from the bucket menu `{15m, 1h, 4h, 12h, 1d, 3d, 7d}`. Off-menu durations permitted in v0.2 via interpolation (§3.4); v0.1 restricts to the menu.
- `R` Coverage Range, `R_min ≤ R ≤ R_max` (initial 0.05–0.50)

2.2 TOSBI determines

```
P_start   = mark price from Position Contract at quote time
P_end     = P_start × (1 − R)          if LONG
          = P_start × (1 + R)          if SHORT
MaxPayout = Q × |P_start − P_end| = A × R
Premium   = Π        (§3)
Fee       = F        (§3.7)
Reserved  = MaxPayout
```

2.3 Payout at settlement

```
LONG:   Payout = min( Q × max(0, P_start − P_settle),  MaxPayout )
SHORT:  Payout = min( Q × max(0, P_settle − P_start),  MaxPayout )

equivalently  Payout = A × min( R, max(0, loss_fraction) )
              LONG:   loss_fraction = (P_start − P_settle) / P_start
              SHORT:  loss_fraction = (P_settle − P_start) / P_start
```

This is a bull put spread (LONG protection) or bear call spread (SHORT protection), held long by the trader.



Payout is linear inside the coverage range and capped outside it. The reserved capital equals the cap.

§3 Pricing

3.1 Black–Scholes building blocks

With risk-free rate `r` (set `r = 0` for MVP; USDC carry is negligible at these tenors) and `S = P_start`:

```
d1(K) = [ ln(S/K) + (r + σ²/2) τ ] / (σ √τ)
d2(K) = d1(K) − σ √τ

Put(K)  = K e^(−rτ) N(−d2) − S N(−d1)
Call(K) = S N(d1) − K e^(−rτ) N(d2)
```

Reference: Black & Scholes (1973); Hull, *Options, Futures and Other Derivatives*, ch. 15.

3.2 Fair premium (per contract)

```
LONG:   Π_fair = Q × [ Put(P_start)  − Put(P_end)  ]      σ = σ_down(asset, τ)
SHORT:  Π_fair = Q × [ Call(P_start) − Call(P_end) ]      σ = σ_up(asset, τ)

sanity check (ATM, r = 0):   Put_ATM = Call_ATM ≈ 0.3989 × S × σ√τ
```

The two directions use different volatilities because crypto returns are asymmetric: falls are sharper and fatter-tailed than rises. With a single σ an ATM put spread and an ATM call spread price identically under Black–Scholes (r = 0), which would misprice one side.

`Put(P_start)` dominates because it is at-the-money at purchase; `Put(P_end)` is small for short tenors but must be computed — it matters for 3–7 day covers and small `R`.

3.3 Volatility table

`σ_dir(asset, τ_bucket)` is maintained off-chain, one value per (asset, bucket, direction): a `σ_down` calibrated on falls and a `σ_up` calibrated on rises.

> **Do not derive short-bucket σ by scaling annualized vol with √τ.** Short-horizon crypto returns are fat-tailed; a √τ-scaled diffusion under-prices 15-minute to 4-hour covers (Matic, Packham & Härdle 2023; Du 2025).

Calibrate each bucket empirically. For bucket `τ_b`, collect rolling `τ_b`-horizon log returns over trailing window `H_b` (initial: 30 days for ≤ 12h buckets, 90 days for ≥ 1d). The **quantile-matched σ** is the σ at which the Black–Scholes spread price equals the empirical expected payoff:

```
down_τ = (P_0 − P_τ) / P_0 = 1 − exp(ret_τ)        loss fraction for a LONG over horizon τ_b
up_τ   = (P_τ − P_0) / P_0 = exp(ret_τ) − 1        loss fraction for a SHORT over horizon τ_b

σ_down,b = argmin_σ | [Put_BS(S,σ,τ_b)  − Put_BS(S(1−R_ref),σ,τ_b) ] / S  −  E_emp[ max(0, mi
σ_up,b   = argmin_σ | [Call_BS(S,σ,τ_b) − Call_BS(S(1+R_ref),σ,τ_b)] / S  −  E_emp[ max(0, mi

both sides as a fraction of A.   R_ref = 0.25 (or solve at 2–3 reference ranges and take the
floor:  σ_dir,b ≥ σ_floor(asset)     initial XLM 60%, BTC/ETH 50%, same floor both directions
cap:    σ_dir,b ≤ 300%
expect σ_down > σ_up for most assets in most regimes; do not force them equal
```

This absorbs jump risk into σ without a separate jump model. Refresh every 4 hours for buckets ≤ 12h, daily for ≥ 1d; re-price immediately on a vol-shock trigger (realized 1h vol > 2× table σ_1h).

v1.1 upgrade: SVI surface

```
k = ln(K/S)
w(k) = a + b [ ρ (k − m) + sqrt( (k − m)² + s² ) ],     σ²τ = w(k)
```

Subject to Gatheral–Jacquier no-arbitrage constraints; calibrated to Deribit/Bybit BTC/ETH surfaces and to empirical skew for XLM. Gives correct `Put(P_end)` at large `R`.

3.4 Off-bucket durations (v0.2)

Interpolate in total variance, not in σ:

```
w_dir,b = σ_dir,b² τ_b          (interpolate each direction separately)
w(τ)  = w_lo + (w_hi − w_lo) × (τ − τ_lo) / (τ_hi − τ_lo)      τ_lo ≤ τ ≤ τ_hi
σ(τ)  = sqrt( w(τ) / τ )

require w(τ) non-decreasing in τ (calendar no-arbitrage); if violated, bump w_hi
```

Long-term path: map `D → q` and price with the Amortizing Perpetual Option closed form (Feinstein 2025), which prices all durations from one formula. Note the provisional US patent on AmPOs before adopting.

3.5 Risk loadings

```
Π = Π_fair × M_u × M_c × (1 + ρ)

utilization:     U'  = (Σ active MaxPayout + MaxPayout_new) / K
                 M_u = 1 + λ_u × U'^p_u            initial λ_u = 0.5, p_u = 2

concentration:   C'  = (Σ MaxPayout(a,d) + MaxPayout_new) / K
                 M_c = 1 + λ_c × C'^p_c            initial λ_c = 1.0, p_c = 2

risk margin:     ρ = 0.05
```

U' / C'	M_u (λ=0.5, p=2)	M_c (λ=1.0, p=2)
0.10	1.005	1.010
0.25 (concentration cap)	1.031	1.063
0.40	1.080	—
0.60	1.180	—
0.90 (utilization cap)	1.405	—

The concentration hard cap (§4.2) binds before `M_c` gets large; that is intentional. Rationale: option market makers charge more as net inventory imbalance grows (Alexander et al. 2021); actuarially these are variance / standard-deviation premium principles (Kaas et al., ch. 5).

3.6 Minimum premium

```
Π = max( Π, Π_min ),      Π_min = max( 0.5 USDC, 0.10% × A )
```

3.7 Protection Fee

```
F = max( F_min, f_bps × A ),     initial F_min = 0.5 USDC, f_bps = 30 bp
```

Fee goes to TOSBI treasury; revenue share with the PerpDEX is a config parameter `share_pdx ∈ [0,1]`, initial 0.

3.8 Total cost to trader

```
Cost = Π + F
Π → ProtectionVault (belongs to capital providers)
F → TOSBI treasury
```

---

§4 Capital rules

4.1 Full backing (invariant, enforced on-chain)

```
Σ_{active} MaxPayout_i  ≤  K_available
K_available = vault_balance − pending_payouts − withdrawal_locked
```

Mint fails if the invariant would be violated.

4.2 Hard caps (enforced on-chain)

Cap	Initial value
Per (asset, direction) concentration `C_{a,d}`	≤ 25% of K
Per asset, both directions	≤ 40% of K
Per trader wallet, sum of active MaxPayout	≤ 2% of K and ≤ 25,000 USDC
Per contract `A_max` / `A_min`	50,000 / 20 USDC
Global utilization `U`	≤ 90% (last 10% is buffer for oracle/settlement slippage)

4.3 Liquidity-relative cap (pricer-enforced, on-chain param)

```
Σ protected notional(a) ≤ κ × Depth_a       initial κ = 0.5
```

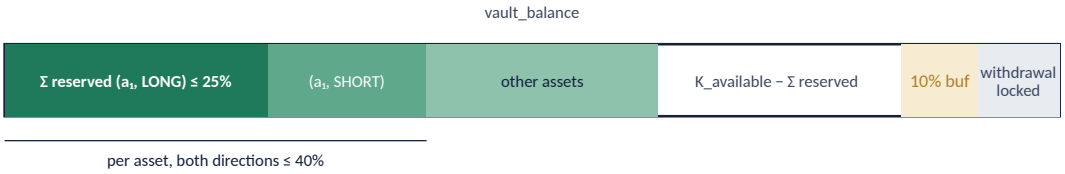
`Depth_a` = USDC needed to move asset `a` by `R_ref` across the venues the oracle reads, estimated daily. A manipulator must spend more than they can extract (Mackinga, Nadahalli & Wattenhofer 2022).

4.4 PerpDEX capital withdrawal

Withdrawals are queued: the requested amount becomes `withdrawal_locked` immediately (blocks new mints against it) and is releasable after `T_withdraw` (initial 7 days) provided the full-backing invariant still holds.

4.5 Premium accrual

Premiums enter the vault immediately and count toward `K`. Track `premium_earned` and `payout_paid` per epoch for loss-ratio monitoring (§11).



All caps are computed against `K`; `withdrawal_locked` is excluded from `K_available` the moment a withdrawal is requested.

§5 Settlement

5.1 Normal settlement (expiry)

```
P_settle = TWAP_oracle( asset, window = W(D), ending at T_expire )

W(D) = clamp( 300 s,  0.53 × sqrt(D_minutes) minutes,  1800 s )
size override:  if MaxPayout > 10,000 USDC then W = 1800 s
```

Duration	W(D)	Duration	W(D)
15m	5 min (floor)	12h	~14 min
1h	5 min (floor)	1d	~20 min
4h	~8 min	3d, 7d	30 min (cap)

Rationale: manipulation cost should scale with contract value. The 30-minute cap matches Bybit Perp Protect and Deribit convention. Settlement may be triggered by anyone at `t ≥ T_expire`; if the keeper is late by more than one oracle resolution, the contract uses the closest available window and emits `LateSettlement`.

5.2 Early settlement (position closed)

```
P_ref = lastprice_oracle(asset) at t_close (or nearest record)
if | P_close − P_ref | / P_ref ≤ δ_close:   P_settle = P_close      δ_close = 1.5%
else:                                       P_settle = P_ref        (do not trust an outlier
```

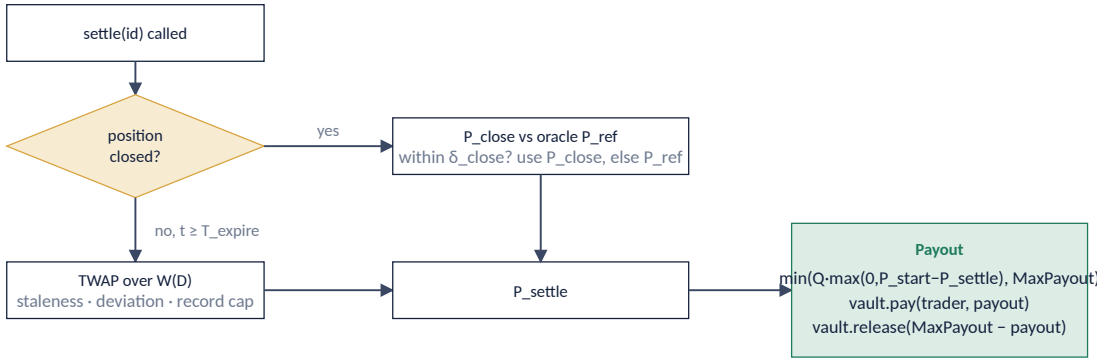
Prevents wash-trading a bad close price on a thin book. Partial close: v0.1 settles the whole contract on any close; v0.2 continues on `Q' = min(Q, remaining size)` with pro-rata `MaxPayout` and reserve release.

5.3 Liquidation of the underlying position

Treated as early settlement with `P_close = liquidation price` from the Position Contract, subject to the same `δ_close` check. Protection survives liquidation (matches Bybit).

5.4 Oracle safety checks (on-chain)

```
staleness:      reject if oracle.last_timestamp < now − 2 × oracle.resolution; retry later
deviation:      if secondary oracle present and |P_primary − P_secondary| / P_secondary > δ_o
                    P_settle = the value closer to the pre-window 1h TWAP
per-record cap: ignore any single record deviating > 15% from the prior record  (truncated-or
stale-max:      if stale > T_stale_max (2h): settle on last valid TWAP, emit DegradedSettleme
```



One settlement path for expiry, early close and liquidation; only the source of `P_settle` differs.

§6 Anti-abuse rules

Rule	Value	Purpose
Position binding	trader must hold open position ≥ `Q`, same asset/direction, at mint	prevents "close the long, keep the put"
Activation delay	protection active at `t_mint + 300 s`; `P_start` fixed at mint	stops buying at the instant a move begins
Quote TTL	signed quote valid 30 s	prevents stale-price arbitrage
Min age of position	60 s since open	discourages open-protect-close loops
One active contract per position per direction	v0.1	simplifies settlement; multiple in v0.2
Event surcharge	pricer applies `M_event ≥ 1` in configured windows (unlocks, macro prints)	informed-flow pricing
Vol-shock repricing	realized 1h vol > 2× table → bump σ, pause ≤ 1h buckets 15 min	avoids selling into a crash at stale prices
Per-wallet cap	§4.2	limits a single informed player

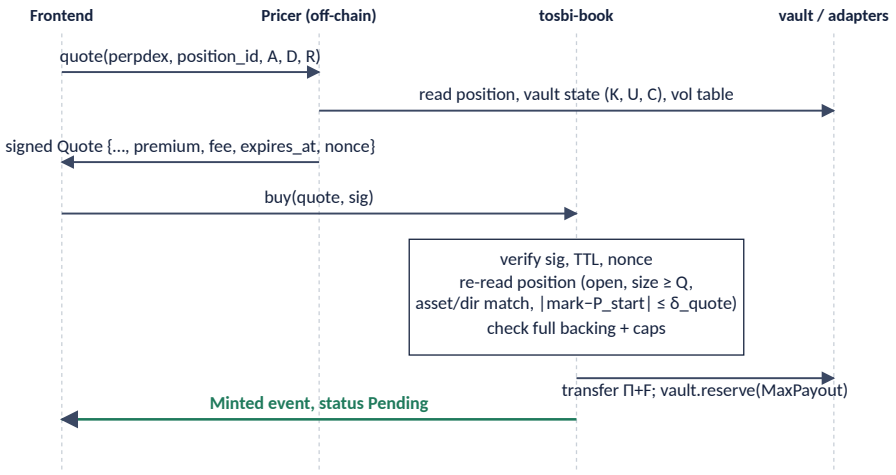
The premium loadings (§3.5) are the primary economic defense; these rules close mechanical loopholes.

§7 Soroban contract architecture

7.1 Contracts

Contract	Responsibility
`tosbi-registry`	params, allowlisted PerpDEX adapters, pause, admin (multisig)
`tosbi-vault`	Protection Capital accounting, reserve/release, payout, withdrawal queue
`tosbi-book`	mint / settle protection contracts, per-contract state
`tosbi-quote-verifier`	verifies TOSBI pricer signature over quote payload
`tosbi-oracle-adapter`	wraps SEP-40 oracle(s): lastprice, twap, decimals, resolution, safety checks
`perpdex-adapter-*`	one per PerpDEX: reads Position Contract, normalizes fields

7.2 Quote and mint flow

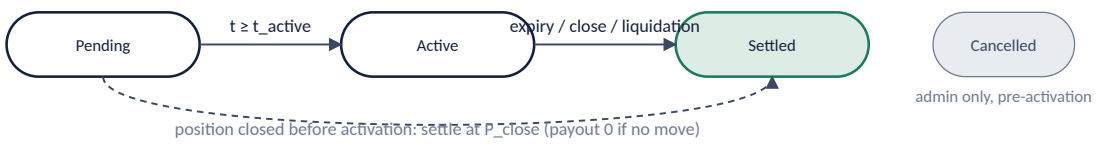


The pricer's loadings may be 30 s stale; the invariant and caps are re-checked on-chain regardless, so a stale quote can never over-commit the vault.

```
Quote {
  perpdex: Address, position_id: BytesN<32>, trader: Address,
  asset: Symbol, direction: u32,
  A: i128, Q: i128, P_start: i128, P_end: i128,
  D: u64, MaxPayout: i128, premium: i128, fee: i128,
  vol_used: u32, quoted_at: u64, expires_at: u64, nonce: u64
}
sig = ed25519( pricer_key, sha256(Quote) )
```

7.3 Contract record (persistent storage)

```
struct Protection {
    id: u64,
    trader: Address,
    perpdex: Address,
    position_id: BytesN<32>,
    asset: Symbol,
    direction: Direction,        // Long | Short
    q: i128,                     // units, 7 decimals
    p_start: i128,               // oracle decimals
    p_end: i128,
    max_payout: i128,            // USDC 7 decimals
    premium_paid: i128,
    fee_paid: i128,
    t_mint: u64,
    t_active: u64,               // t_mint + activation delay
    t_expire: u64,
    window_secs: u32,
    status: Status,              // Pending | Active | Settled | Cancelled
    settle_price: Option<i128>,
    payout: Option<i128>,
}
```



Status transitions. There is no trader-initiated cancel; `Cancelled` exists only for admin recovery before activation.

7.4 Storage and TTL

- `Protection` records, vault balances, reserve totals, per-(asset,direction) totals, per-wallet totals: **persistent** storage.
- Nonce set: persistent, pruned after settlement. Quote payloads: not stored.
- Extend TTL on every write and on every read in `settle`; keeper extends TTL for all active records every 24h. Target TTL ≥ `D + 14 days`.
- Never use temporary storage for anything that affects money.

7.5 Keeper

Off-chain service that (a) calls `settle(id)` at `t_expire` within one oracle resolution; (b) watches PerpDEX close events and calls `settle_early(id)`; (c) extends TTLs; (d) posts vol-table hashes. Anyone may call `settle`; the keeper is a liveness convenience, not a trust dependency.

7.6 Required PerpDEX Position Contract interface

```
fn get_position(position_id: BytesN<32>) -> Position {
    owner: Address, asset: Symbol, direction: Direction,
    size: i128, entry_price: i128, mark_price: i128,
    status: PositionStatus,          // Open | Closed | Liquidated
    close_price: Option<i128>, close_ts: Option<u64>,
}
event PositionClosed(position_id, close_price, close_ts, reason)
```

If a PerpDEX cannot expose `close_price` on-chain, the adapter falls back to oracle-only early settlement (`P_settle = P_ref`).

7.7 Oracle adapter (SEP-40)

Uses `lastprice(asset)`, `twap(asset, records)`, `decimals()`, `resolution()`. `records = ceil(W / resolution)`. Read `decimals()` at runtime; never hard-code. Primary: Reflector. Secondary (deviation check only): Pyth or RedStone on Stellar.

> If Reflector's public resolution is 5 minutes, the 5-minute floor equals one record. For ≤ 1h buckets either use Reflector's paid faster feed or accept single-record settlement with the per-record cap.

7.8 Events

`Minted`, `Activated`, `Settled{price, payout}`, `SettledEarly`, `LateSettlement`, `DegradedSettlement`, `CapitalDeposited`, `WithdrawalRequested`, `WithdrawalExecuted`, `ParamsUpdated`, `Paused`.

7.9 Admin and upgrades

Multisig admin (TOSBI + PerpDEX signers). Parameter changes time-locked 24h except `pause`. Contract upgrade via Soroban WASM upgrade behind the same time-lock.

§8 Off-chain services

Service	Responsibility
Pricer	vol table, loadings, quote signing; stateless HTTP; signing key in HSM/KMS
Vol calibrator	empirical bucket σ (§3.3), SVI (v1.1), event calendar, vol-shock detection
Keeper	settlement, early settlement, TTL extension
Risk monitor	utilization, concentration, loss ratio, oracle health, alerts; can trigger pause
Depth estimator	daily `Depth_a` for the liquidity cap (§4.3)

Publish the vol table and its hash on-chain (or IPFS + on-chain hash) per refresh so PerpDEX partners can audit quotes.

§9 Initial risk parameters

Parameter	Value	Section	Parameter	Value	Section
Buckets	15m, 1h, 4h, 12h, 1d, 3d, 7d	2.1	Concentration cap	25% per (asset,dir); 40% per asset	4.2
`R_min`, `R_max`	5%, 50%	2.1	Wallet cap	2% of K, ≤ 25k USDC	4.2
`r`	0	3.1	`A_min`, `A_max`	20, 50,000 USDC	4.2
`σ_floor`	XLM 60%, BTC/ETH 50%, both directions	3.3	Global `U` cap	90%	4.2
`λ_u`, `p_u`	0.5, 2	3.5	`κ`	0.5	4.3
`λ_c`, `p_c`	1.0, 2	3.5	`T_withdraw`	7 days	4.4
`ρ`	5%	3.5	`W(D)`	clamp(5m, 0.53√D_min, 30m)	5.1
`Π_min`	max(0.5, 0.10% A)	3.6	`δ_close`	1.5%	5.2
`F`	max(0.5, 30 bp × A)	3.7	`δ_oracle`	2%	5.4
Activation delay	300 s	6	`δ_quote`	0.5%	7.2
Quote TTL	30 s	6	`T_stale_max`	2 h	5.5

KPIs to tune these: loss ratio (§11), utilization distribution, quote-to-mint conversion.

---

§10 Worked example

10.1 LONG protection

XLM LONG, `P_start = 0.20`, `A = 1,000 USDC` → `Q = 5,000 XLM`, `D = 1d`, `R = 25%` → `P_end = 0.15`, `MaxPayout = 250`. Vol table: `σ_down,1d(XLM) = 100%`.

```
τ = 1/365.25 = 0.002738          σ√τ = 0.05232

ATM put (r = 0):   d1 = σ√τ/2 = 0.02616,   d2 = −0.02616
Put(0.20) = 0.20 × [ N(0.02616) − N(−0.02616) ] = 0.20 × 0.02087 = 0.004174 per unit
Put(0.15): ln(0.20/0.15) = 0.2877,  d2 ≈ 5.47,  N(−5.47) ≈ 2×10⁻⁸  → negligible

Π_fair = 5,000 × 0.004174 = 20.87 USDC   (2.09% of A)

vault: K = 500,000; active reserve 249,750 → U' = 0.50 → M_u = 1.125
       XLM-long reserve 99,750           → C' = 0.20 → M_c = 1.04   (below 25% cap, §4.2)

Π = 20.87 × 1.125 × 1.04 × 1.05 = 25.64 USDC
F = max(0.5, 0.003 × 1,000)    =  3.00 USDC
Total cost                      = 28.64 USDC  (2.86% of A)

W = 0.53 × √1440 ≈ 20 min
```

Outcome at expiry	Payout	Note
`P_settle = 0.23`	0	trader net −28.64
`P_settle = 0.17`	150	5,000 × 0.03; net −28.64 after offsetting a 150 position loss
`P_settle = 0.10`	250	5,000 × 0.10 = 500, capped
early close at 0.18, oracle 0.181	100	dev 0.55% ≤ δ_close → use P_close

Vault economics for this contract: +25.64 premium in, 250 reserved for one day, expected payout ≈ 20.87 by construction of σ → expected margin ≈ 4.77 plus the loadings' cushion.

10.2 SHORT protection

XLM SHORT, `P_start = 0.20`, `A = 1,000 USDC` → `Q = 5,000 XLM`, `D = 1d`, `R = 25%` → `P_end = 0.25`, `MaxPayout = 250`. Vol table: `σ_up,1d(XLM) = 90%` (calibrated on rises; lower than the 100% downside figure).

```
σ√τ = 0.90 × 0.05232 = 0.04709

ATM call (r = 0):  d1 = σ√τ/2 = 0.02355,   d2 = −0.02355
Call(0.20) = 0.20 × [ N(0.02355) − N(−0.02355) ] = 0.20 × 0.01879 = 0.003757 per unit
Call(0.25): ln(0.20/0.25) = −0.2231,  d1 ≈ −4.71,  N(−4.71) ≈ 1.2×10⁻⁶  → negligible

Π_fair = 5,000 × 0.003757 = 18.79 USDC   (1.88% of A)

vault: K = 500,000; active reserve 249,750 → U' = 0.50 → M_u = 1.125
       XLM-short reserve 49,750          → C' = 0.10 → M_c = 1.01

Π = 18.79 × 1.125 × 1.01 × 1.05 = 22.41 USDC
F                               =  3.00 USDC
Total cost                      = 25.41 USDC  (2.54% of A)
```

Outcome at expiry	Payout	Note
`P_settle = 0.17`	0	position gained 150; trader net −25.41
`P_settle = 0.23`	150	5,000 × 0.03; net −25.41 after offsetting a 150 position loss
`P_settle = 0.30`	250	5,000 × 0.10 = 500, capped
early close at 0.22	100	5,000 × 0.02

The short costs less than the long on the same terms (25.41 vs 28.64) for two reasons: the upside vol is lower, and short protection is less crowded in this vault state, so its concentration loading is smaller.

§11 Monitoring

```
LR_epoch = Σ payouts / Σ premiums       epoch = 7 days; also 30-day rolling

target:   LR ≤ 0.80 rolling 30d
raise ρ or bucket σ:   LR > 1.0 for two consecutive weeks
lower ρ / loadings:    LR < 0.4 for 8 weeks with low utilization
```

Track LR per bucket and per asset to see which tenors are mispriced; expect short buckets to fail first. Also: utilization time series; max concentration; oracle staleness incidents; settlement latency; count of `δ_close` overrides (wash-trading signal); quote-to-mint conversion by price level (demand elasticity).

§12 Test and validation plan

Pricing backtest
XLM/BTC/ETH hourly, 2022–2026. Simulate selling every bucket × R daily; compute realized LR under the empirical-σ calibration; confirm LR < 1 over the full period; inspect worst 30-day windows (May 2022, Nov 2022, Aug 2024, Oct 2025, Feb 2026).

Crash stress

Instantaneous −30% and −50% with a fully concentrated book at cap; verify solvency and that payouts equal Σ MaxPayout.

Oracle attack simulation
Inject single-record spikes of +15% / −15% / −40% into the TWAP window; confirm per-record cap and deviation logic bound payout error to < δ_oracle.

Wash-close test
Close position at ±3% from oracle; verify δ_close fallback.

Invariant fuzzing
Randomized mint/settle/withdraw sequences; assert full-backing and cap invariants after every step.

TTL test
7-day contract with no interaction; verify keeper extension and archival-restore path.

Latency
Keeper settles within one oracle resolution of expiry ≥ 99% of the time on testnet.

Audit
Apply to the Soroban Audit Bank via SCF Build.

§13 Roadmap after v0.1

Version	Change
v0.2	Off-bucket durations (variance interpolation), partial close, multiple contracts per position
v1.1	SVI surface; secondary oracle deviation check live; event calendar surcharges
v2	Tranched vault: PerpDEX first-loss junior + public senior (Architecture 4)
v2.x	Streaming / AmPO product for open-ended protection (Architecture 3); resolve patent question first
Later	Partial delta-hedging once a Stellar perp venue has deep mainnet liquidity (Architecture 5)

§14 References

Option pricing and volatility

- Black, F. & Scholes, M. (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy* 81(3).
- Hull, J. C. *Options, Futures, and Other Derivatives* (11th ed.), ch. 12, 15.
- Gatheral, J. & Jacquier, A. (2014). Arbitrage-free SVI volatility surfaces. *Quantitative Finance* 14(1). arxiv.org/abs/1204.0646
- Du, L. (2025). Pricing Cryptocurrency Options With Volatility of Volatility. *Journal of Futures Markets*. doi:10.1002/fut.70029
- Matic, J. L., Packham, N. & Härdle, W. K. (2023). Hedging Cryptocurrency Options. *Review of Derivatives Research* 26(1). arxiv.org/abs/2112.06807
- Alexander, C. & Imeraj, A. (2023). Delta hedging bitcoin options with a smile. *Quantitative Finance* 23(5). tandfonline.com

- Alexander, C., Deng, J., Feng, J. & Wan, H. (2021). Net Buying Pressure and the Information in Bitcoin Option Trades. arxiv.org/abs/2109.02776
- Zbandut, A. (2025). Pricing of wrapped Bitcoin and Ethereum on-chain options. arxiv.org/abs/2512.20190

Perpetual and streaming designs (§3.4, roadmap)

- Feinstein, Z. (2025). Amortizing Perpetual Options. arxiv.org/abs/2512.06505
- Bichuch, M. & Feinstein, Z. (2026). Designing On-Chain Options: Amortizing Perpetual Options. arxiv.org/abs/2605.19146
- Lambert, G. & Kristensen, J. (2022). Panoptic: the perpetual, oracle-free options protocol. arxiv.org/abs/2204.14232
- Ackerer, D., Hugonnier, J. & Jermann, U. (2026). Perpetual Futures Pricing. *Mathematical Finance*. arxiv.org/abs/2310.11771
- Angeris, G., Chitra, T., Evans, A. & Lorig, M. (2023). A primer on perpetuals. *SIAM J. Financial Mathematics* 14(1).

Insurance capital and pricing

- Kaas, R., Goovaerts, M., Dhaene, J. & Denuit, M. *Modern Actuarial Risk Theory* (2nd ed.), ch. 5.
- Nadler, M., Bekemeier, F. & Schär, F. (2022). DeFi Risk Transfer: Towards A Fully Decentralized Insurance Protocol. arxiv.org/abs/2212.10308
- Singh, S. F., Michalopoulos, P. & Veneris, A. (2024). BAKUP: Automated, Flexible, and Capital-Efficient Insurance Protocol for DeFi. arxiv.org/abs/2410.09341
- Bekemeier, F., Schär, F. & Schmeiser, H. (2025). Decentralized Finance risk transfer and smart contract-based insurance. *Journal of Banking & Finance*. sciencedirect.com

Oracle manipulation and settlement

- Mackinga, T., Nadahalli, T. & Wattenhofer, R. (2022). TWAP Oracle Attacks: Easier Done than Said? IACR ePrint 2022/445. eprint.iacr.org/2022/445
- Chaos Labs. Uniswap V3 TWAP: Assessing TWAP Market Risk. chaoslabs.xyz
- Hacken. Uniswap v4 Truncated Oracle. hacken.io

Industry references

- Bybit Perp Protect. help center, guide
- Nami Insurance docs. docs.namiinsurance.io
- Bumper. bumper.fi

Stellar / Soroban

- SEP-40 Oracle interface. github.com/stellar/stellar-protocol
- Reflector Network. docs, contract
- Stellar oracle providers. developers.stellar.org
- Soroban state archival / TTL. developers.stellar.org
- Stellar Community Fund. stellar.org/grants-and-funding
- Circle USDC on Stellar. circle.com

TOSBI · Architecture 1 technical specification · v0.1 draft · Companion documents: "How it works" (explainer) and "Business model".